

Project 5 – Secure Shell (SSH)

Contents

Background	1
Objectives	1
Prerequisites	1
Using Windows 10 SSH Client.....	2
Using Fedora Server.....	5
Go back to your Windows machine.....	7
Submission.....	10

Background

Explore the use of SSH on Windows and Linux platforms using the command line.

Objectives

1. Use respective native SSH client to connect to Fedora server.
2. Highlight additional features of SSH clients.

Prerequisites

1. A Windows or MacOS device with a functioning hypervisor (e.g., VMware Workstation Pro or VMware Fusion Pro).

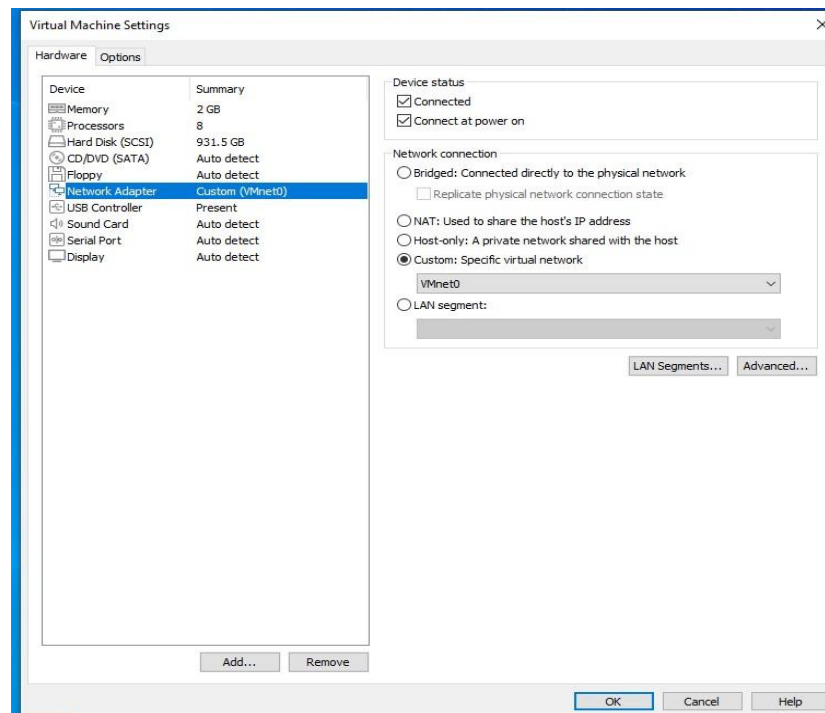


Using Windows 10 SSH Client

1. Start VMware Workstaion Pro.
2. Copy the WIN-10 virtual machine to your folder in the V drive.
3. Power on this virtual machine.

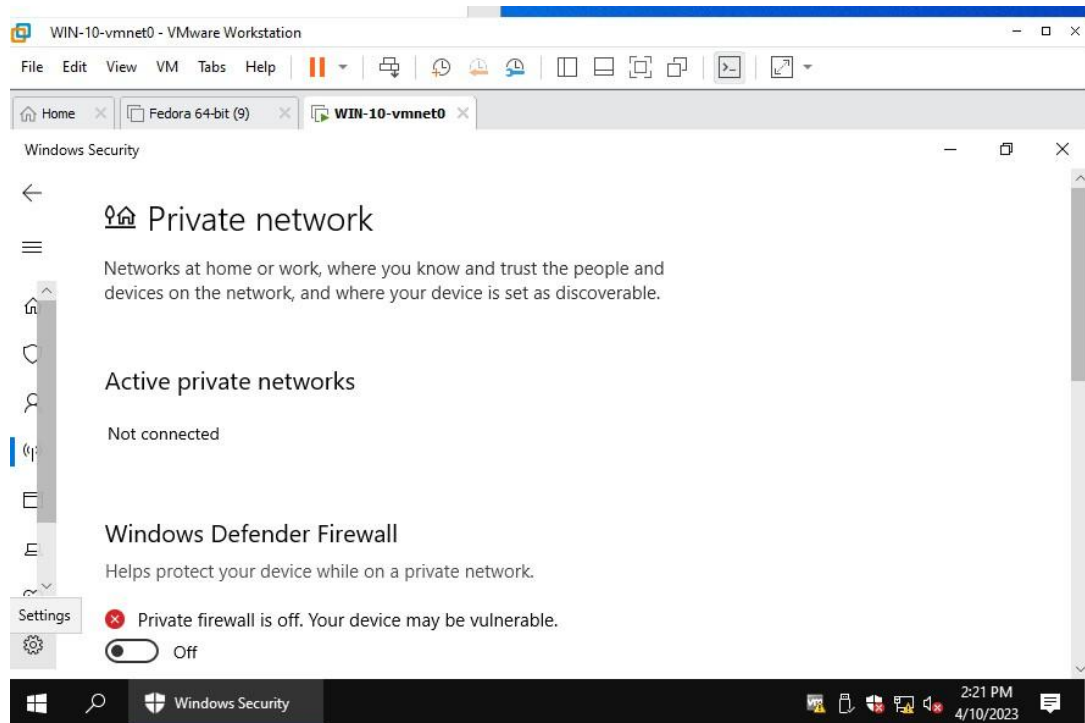


- 4 Make sure it is connected to the vmnet0 network. Go to VM and **settings to change** the network.

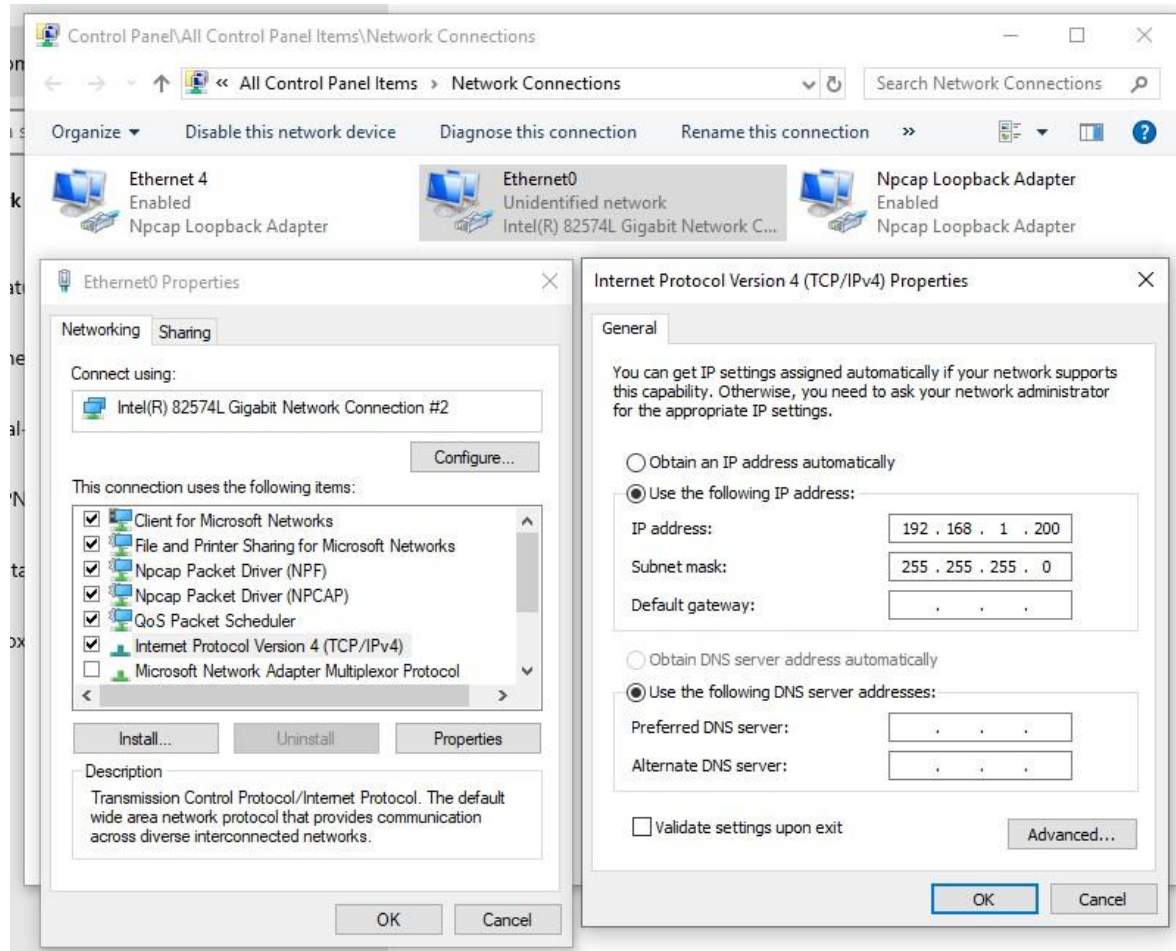




2. On the right bottom side of the windows virtual machine screen click on **firewall**. **Turn off** all the firewalls.



3. Next is to select the **internet and network settings**, click on **ethernet4**, then click on change adapter options.
4. Click on **ethernet0**.
5. Click on **properties** and then double click on **Internet protocol version 4**.
6. Type **IP 192.168.1.200** and subnet mask **255.255.255.0**



7. Type cmd in the search bar.
8. Now turn on the fedora machine.



Using Fedora Server

1. Start Fedora Server virtual machine.
2. Login as **root**.

3. Make sure your **fedora server** is on **vmnet0** as well.

4. Type **ifconfig** to see if you have an IP.

5. Set the IP using the following command.

```
ifconfig -v ens160 192.168.1.200 netmask 255.255.255.0
```

6. Check the firewall status-

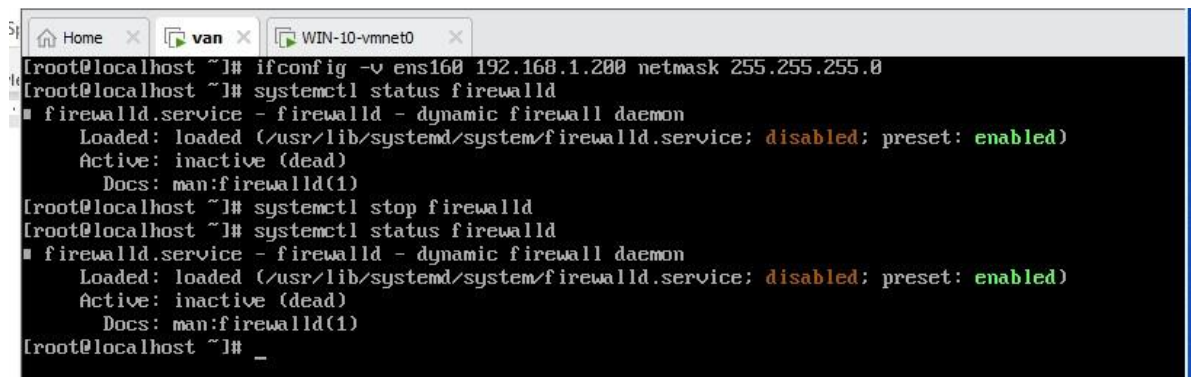
```
systemctl status firewalld
```

7. Stop the firewall if active

```
systemctl stop firewalld
```

8. Check again to make sure its deactivated.

```
systemctl status firewalld
```

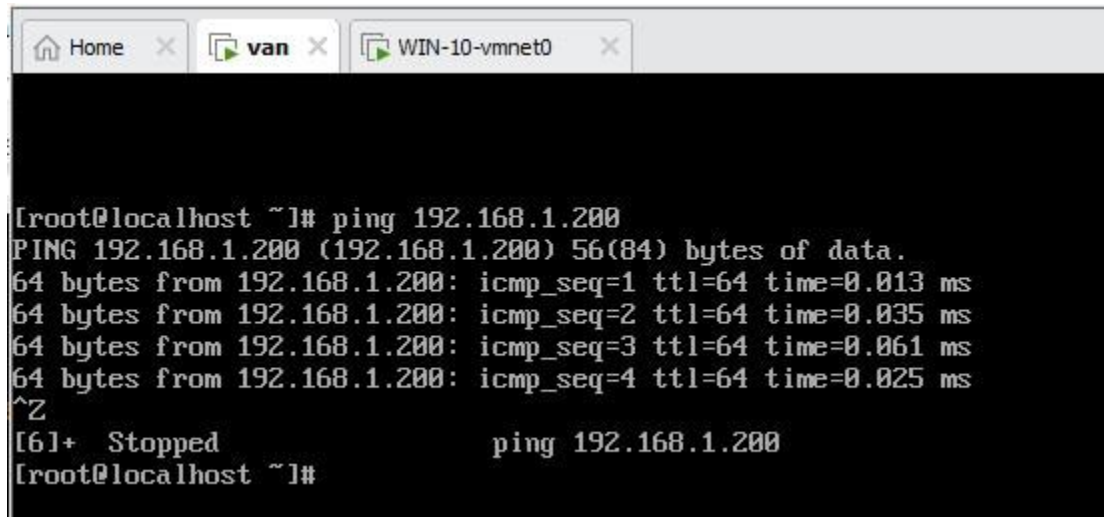


```
[root@localhost ~]# ifconfig -v ens160 192.168.1.200 netmask 255.255.255.0
[root@localhost ~]# systemctl status firewalld
■ firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: man:firewalld(1)
[root@localhost ~]# systemctl stop firewalld
[root@localhost ~]# systemctl status firewalld
■ firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: man:firewalld(1)
[root@localhost ~]# _
```



9. Type **ping 192.168.1.200** (or whatever IP address your Windows machine has.)

You should be able to ping.



```
Home x van x WIN-10-vmnet0 x
[root@localhost ~]# ping 192.168.1.200
PING 192.168.1.200 (192.168.1.200) 56(84) bytes of data:
64 bytes from 192.168.1.200: icmp_seq=1 ttl=64 time=0.013 ms
64 bytes from 192.168.1.200: icmp_seq=2 ttl=64 time=0.035 ms
64 bytes from 192.168.1.200: icmp_seq=3 ttl=64 time=0.061 ms
64 bytes from 192.168.1.200: icmp_seq=4 ttl=64 time=0.025 ms
^Z
[6]+  Stopped                  ping 192.168.1.200
[root@localhost ~]#
```

Go back to your Windows machine

1. Ping the IP address or hostname of the Fedora server to ensure network connectivity.

ping 192.168.1.200

you should be able to ping.

Van Nguyen

CET479 – Network Administration

2. Run the following command:

```
ssh username@hostname/ip
```



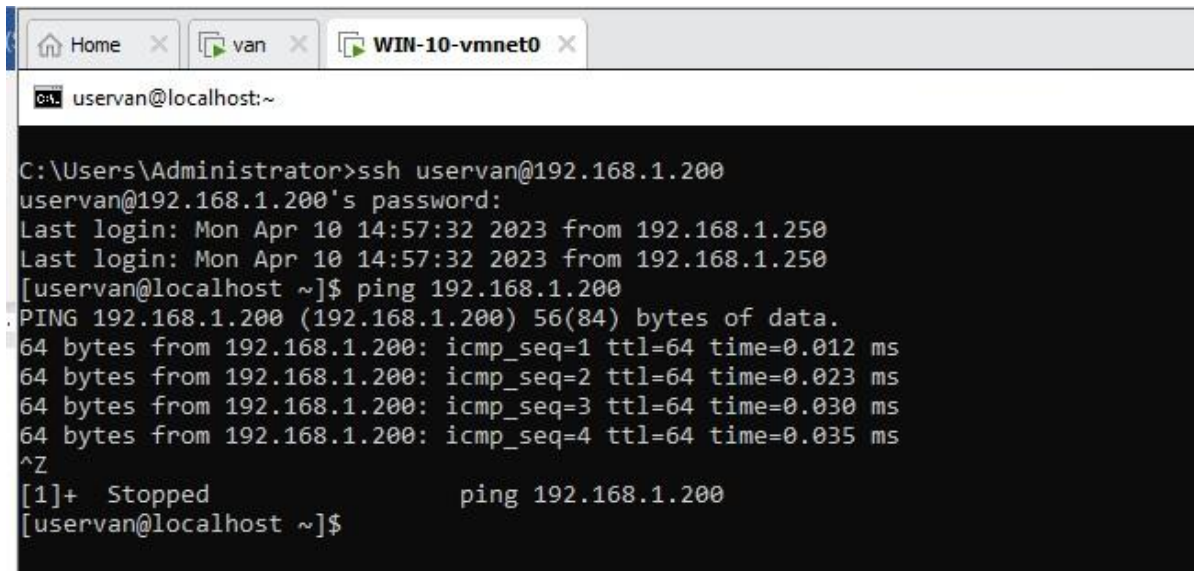
```
Command Prompt - ssh sorgiodo@192.168.1.96

C:\Users\Nick Sorgio>ssh sorgiodo@192.168.1.96
The authenticity of host '192.168.1.96 (192.168.1.96)' can't be established.
ECDSA key fingerprint is SHA256:fugWIIHKEpMSbBjHpk2hvn+hTZsw25I2vihauTnN6mM.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

In your case the command would be

```
ssh username@192.168.1.200
```

3. Enter appropriate password.



```
C:\Users\Administrator>ssh uservan@192.168.1.200
uservan@192.168.1.200's password:
Last login: Mon Apr 10 14:57:32 2023 from 192.168.1.250
Last login: Mon Apr 10 14:57:32 2023 from 192.168.1.250
[uservan@localhost ~]$ ping 192.168.1.200
PING 192.168.1.200 (192.168.1.200) 56(84) bytes of data.
64 bytes from 192.168.1.200: icmp_seq=1 ttl=64 time=0.012 ms
64 bytes from 192.168.1.200: icmp_seq=2 ttl=64 time=0.023 ms
64 bytes from 192.168.1.200: icmp_seq=3 ttl=64 time=0.030 ms
64 bytes from 192.168.1.200: icmp_seq=4 ttl=64 time=0.035 ms
^Z
[1]+  Stopped                  ping 192.168.1.200
[uservan@localhost ~]$
```



4. Run the **whoami**, **pwd**, **hostname**, and **ifconfig** commands. You should be able to remotely control your *fedora machine*.

```
[uservan@localhost ~]$ whoami
uservan
[uservan@localhost ~]$ pwd
/home/uservan
[uservan@localhost ~]$ ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.200 netmask 255.255.255.0 broadcast 192.168.1.255
    ether 00:0c:29:69:8e:d4 txqueuelen 1000 (Ethernet)
    RX packets 407 bytes 36317 (35.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 283 bytes 30100 (29.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 127 bytes 11536 (11.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 127 bytes 11536 (11.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[uservan@localhost ~]$
```

Submission

Use this document (or one of your choosing) to submit your screen shots and any other relevant data. At a minimum, you should show the commands run after logging in as root. Please ensure your submission includes your name, class, date, and reference to this lab.

